Study Guide

11.1 Linear Systems: Theory

- Review of linear algebra

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\boxed{\text{tr } A = a+d = \tau}$

$\boxed{\det A = ad - bc = \Delta}$

eigenvalues: $\boxed{\lambda^2 - \tau\lambda + \Delta = 0}$

i) Stability analysis

$\Delta = \lambda_1 \lambda_2$

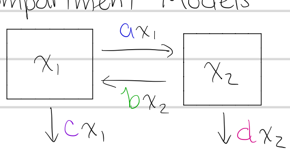
$\tau = \lambda_1 + \lambda_2$

$\lambda = \dfrac{\tau \pm \sqrt{\boxed{\tau^2 - 4\Delta}}}{2}$ ⟵ $\tau^2 - 4\Delta$ determines if eigenvalues are real or complex

Calculate Δ
- (−): λ_1, λ_2 are real and mixed signs — saddle point
- (0): inconclusive
- (+): Calculate $\tau^2 - 4\Delta$
  - (−): λ_1, λ_2 are imaginary. Look at τ
    - (−): $Re(\lambda) < 0$ — stable spiral
    - (0): $Re(\lambda) = 0$ — center
    - (+): $Re(\lambda) > 0$ — unstable spiral
  - (+): λ_1, λ_2 are real. Look at τ
    - (−): $\lambda_1, \lambda_2 < 0$ — stable node
    - (+): $\lambda_1, \lambda_2 > 0$ — unstable node

11.2 Linear Systems: Applications

i) Compartment Models



$\dfrac{dx}{dt} = Ax$ $A = \begin{bmatrix} -(a+c) & b \\ a & -(b+d) \end{bmatrix}$ $\Delta = ad + bc + cd$

eigenvalues are always real

11.3. Nonlinear Systems: Theory

i) **Critical point (equilibrium)** $f(\hat{x}) = 0$

ii) $A = Df(\hat{x})$

iii) **Zero isoclines (nullclines)**

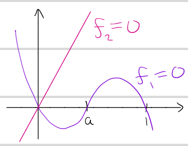
$f_1(x_1, x_2) = 0$

$f_2(x_1, x_2) = 0$

11.4.4 A Mathematical Model for Neuron Activity

1) $\frac{dV}{dt} = -V(V-a)(V-1) - w = f_1(V, w)$

$\frac{dw}{dt} = b(V - cw) = f_2(V, w)$



action potential when $\frac{dV}{dt}(0) > 0$

If $w(0) = 0$, then $a < V(0) < 1$ for action potential.

12.1. Counting

i) **Multiplication Principle** $n_1 n_2 \cdots n_m$

ii) **Permutation** (ordered) $P(n, k) = \frac{n!}{(n-k)!}$

iii) **Combination** (unordered) $C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

iv) Binomial Theorem $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$

v) Multinomial $n = n_1 + n_2 + \cdots + n_m$ $\binom{n}{n_1, n_2, \ldots, n_m} = \frac{n!}{n_1! n_2! \cdots n_m!}$ (Arrange letters or groups)

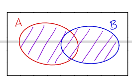
12.2. What is Probability?

i) **Random experiment** is a repeatable experiment in which the outcome is uncertain.

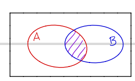
ii) **Sample space** is the set of all possible outcomes. **Example**: Two coin tosses $\Omega = \{HH, HT, TH, TT\}$

iii) **Union** $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$ and **Intersection** $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

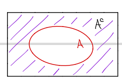
iv) **Venn Diagrams** $A \cup B$



$A \cap B$



iv) **Complement** $A^c = \{x \mid x \notin A\}$



v) **De Morgan's Laws** $(A \cup B)^c = A^c \cap B^c$

$(A \cap B)^c = A^c \cup B^c$

vi) $A_1, A_2, \ldots, A_n$ are **pairwise disjoint** if $A_i \cap A_j = \phi$ whenever $i \neq j$

vii) **Probability** 1. For any event A, $0 \leq P(A) \leq 1$

2. $P(\phi) = 0$ and $P(\Omega) = 1$

3. For any two disjoint events A and B, $P(A \cup B) = P(A) + P(B)$

12.3. Conditional Probability and Independence

i) Conditional probability of A given B: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

ii) **Partition** is the collection of disjoint sets $B_1, B_2, \ldots, B$ such that $\Omega = \bigcup_{i=1}^{n} B_i$

iii) **Law of Total Probability** $P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i)$

iv) Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$

v) **Pairwise Independent** $A_1, A_2, \ldots, A_n$ if $P(A_i \cap A_j) = P(A_i)P(A_j)$ whenever $i \neq j$

vi) **The Bayes Formula** Let $B_1, B_2, \ldots, B_n$ form a partition of Ω, and let A be an event. Then

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}$$